

Electro Pi

Backend Node.js Developer

Technical Assessment Task

Experience Level: 1 – 2 Years

Task Overview

Build a RESTful API for a Project & Task Management System using Node.js. This task assesses your backend development skills, code organization, and understanding of common Node.js patterns.

Required Tech Stack

Category	Technology
Runtime	Node.js (v18+)
Framework	Express.js or Fastify
Database	PostgreSQL or MongoDB
ORM / ODM	Sequelize / TypeORM (PostgreSQL) or Mongoose (MongoDB)
Authentication	JWT (JSON Web Tokens)
Language	JavaScript or TypeScript (TypeScript preferred)
API Docs	Swagger / OpenAPI or Postman Collection

Project Description

Build a Project & Task Management API with the following features:

1. Authentication & Authorization

- User registration (Name, Email, Password)
- User login – return JWT access token
- All other endpoints must require a valid JWT

2. Projects

- Create a new project (Title, Description, Status)
- Get all projects for the authenticated user
- Get a single project by ID
- Update project details
- Delete a project

3. Tasks

- Create a task under a project (Title, Description, Status, Priority, Due Date)
- Get all tasks for a specific project
- Get a single task by ID
- Update task (including status: Pending / In Progress / Done)
- Delete a task
- Filter tasks by status or priority

Technical Requirements

- Use layered architecture (routes → controllers → services → models)
- Validate all inputs (use Joi, Zod, or express-validator)
- Proper error handling with correct HTTP status codes
- Hash passwords using bcrypt
- Database migrations / seed files included
- Clear README.md with setup instructions and .env.example

Bonus Points (Optional)

- Pagination & sorting on list endpoints
- Role-based access control (Admin / Member)
- Unit tests (Jest or Mocha)
- Docker Compose setup
- Use TypeScript

Submission Requirements

Please submit the following:

1. Public GitHub Repository link
2. README.md with: project description, tech stack, how to run locally, .env.example
3. Database migration / seed files
4. Postman Collection (.json) OR Swagger documentation URL
5. Any important notes about your implementation

Evaluation Criteria

Criteria	Weight
Code structure & architecture	25%
API functionality & correctness	25%
Authentication & security	20%
Error handling & validation	15%
Documentation & README	15%

Deadline: 5 working days from receiving this task. Good luck!